CXR Kubernetes Deployment Manual

Bootcamp workflow: Terraform, kind, Helm, and Argo CD

CXR Ops Lab — `cxr-ops-lab`

2026-05-28

Abstract

This manual documents the end-to-end Kubernetes deployment path used for the CXR bootcamp: local **kind** cluster `cxr-lab`, Docker image `cxr-ui:local`, Helm chart `helm/cxr-ui`, optional Terraform provisioning, and Argo CD GitOps from **GitHub** (`UdonsiKalu/cxr-ops-lab`). It explains rationale, folder layout, commands, ports, CI vs. CD, and how to adapt the pattern to another environment.

Contents

Chapter 1

Executive summary

1.1 What you built

- A **local Kubernetes lab** (`kind` / `cxr-lab`) running the rehearsal CXR UI.
- **Helm (SW.4)** packages the app as chart `cxr-ui`.
- **Terraform (SW.5)** can reproducibly ensure the kind cluster exists.
- **Argo CD (SW.8)** is **continuous delivery**: Git on GitHub is the source of truth; Argo syncs the Helm chart into the cluster.
- **CI** (build/test/Trivy) lives in `cxr-ui-rehearsal` — it does **not** deploy to this kind cluster.

1.2 One-line flow

Build image → Load into kind → Helm deploy → Argo keeps Git = cluster → Browser via port-forward

1.3 URLs (lab)

URL	Meaning
`http://localhost:8251`	Daily dev (full analyze)
`http://localhost:3000`	Docker Compose lab (SW.2)
`http://localhost:8081`	K8 UI (port-forward to pod :3000)
`https://localhost:8083`	Argo CD UI

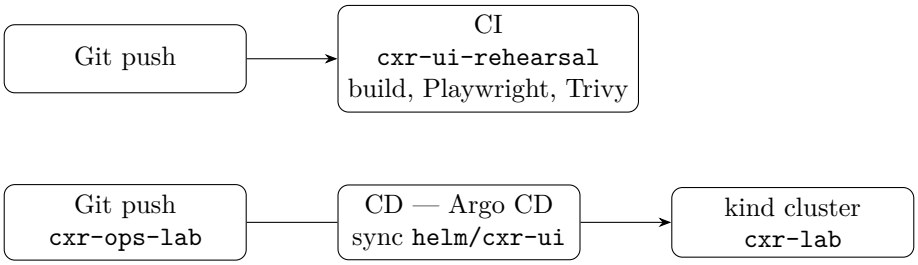
Chapter 2

Rationale: why each layer exists

2.1 Design principles

1. **Do not touch production** `cxrlabs-dev/platform/infra`.
2. **Same app image path** as SW.1 Docker: rehearsal `cxr-ui` built into `cxr-ui:local`.
3. **Learn K8 mechanics** (namespace, deployment, service) before GitOps polish.
4. **Helm** parameterizes the chart (image tag, replicas, env) for SW.4.
5. **Terraform** documents reproducible cluster creation (SW.5 ADR: local kind only for bootcamp).
6. **Argo CD** implements SW.8 GitOps: push chart changes ⇒ cluster reconciles.

2.2 CI vs. CD (critical distinction)



Push-based CD (GitHub Actions runs `helm upgrade`) is a valid alternative but **not wired** for this home lab because the cluster is local, the image is `cxr-ui:local` loaded via `kind load`, and the syllabus target for CD is Argo.

Chapter 3

Repository and folder map

3.1 `cxr-ops-lab` layout (canonical)

```
cxr-ops-lab/
  Dockerfile              # SW.1 image (cxr-ui:local)
  Dockerfile.compose      # SW.2 (not used in K8 pod by default)
  helm/cxr-ui/            # SW.4 Helm chart (Argo source path)
    Chart.yaml
    values.yaml
    templates/
  k8s/                    # SW.3 raw manifests (study / --raw)
    namespace.yaml
    deployment.yaml
    service.yaml
    argocd/
      application-cxr-ui.yaml
  terraform/              # SW.5 kind provisioner
  scripts/
    00-install-tools.sh   # kind, kubectl, helm, terraform, argocd
    01-kind-cluster.sh
    02-build-and-load.sh
    03-k8-up.sh           # one-shot cluster+image+helm
    05-helm-install.sh
    12-k8-ensure.sh
    13-argo-install.sh
    14-stack-test.sh
    15-e2e-deploy.sh      # full stack
  evidence/               # SW.3, SW.4, SW.8 proof
  docs/                   # this manual
  bin/                    # downloaded CLIs (gitignored)
```

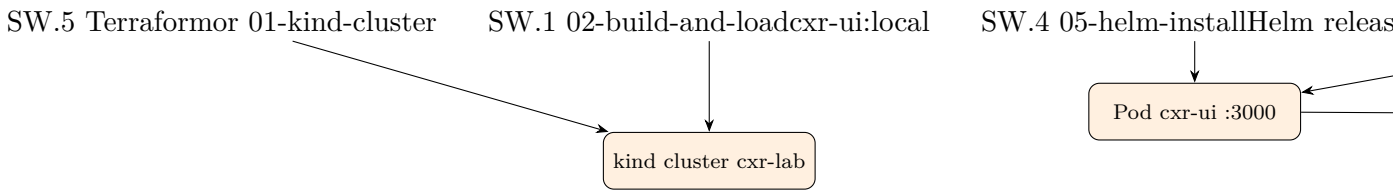
3.2 Related repos (not merged)

- **App source:** `cxr-ui-prune-rehearsal/cxr-ui` (build context for Dockerfile).
- **CI:** `https://github.com/UdonsiKalu/cxr-ui-rehearsal`.
- **Ops + Helm + Argo:** `https://github.com/UdonsiKalu/cxr-ops-lab` (private).
- **Production (untouched):** `cxrlabs-dev/claim_analysis_tools/platform/infra`.

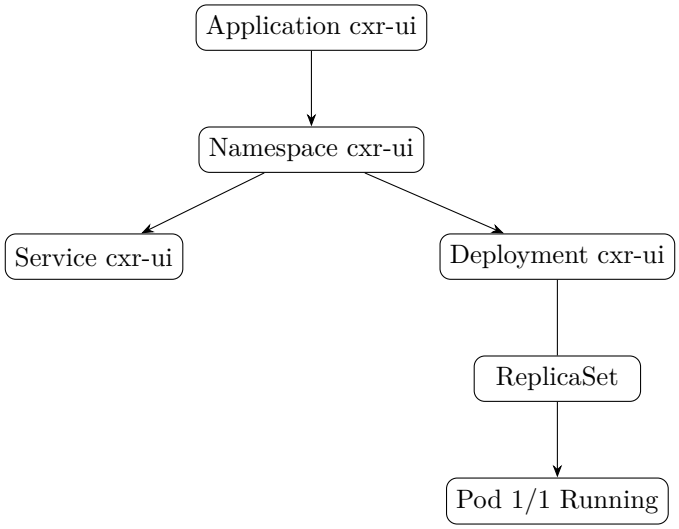
Chapter 4

Architecture diagrams

4.1 End-to-end deployment flow



4.2 In-cluster object graph (Argo tree)



4.3 Dependencies (M4.8): out of cluster

SQL, Qdrant, and Python analyzers stay on the host or Compose stack in the bootcamp K8 pod. Full Claim Studio analyze uses `:8251` or `:3000`, not the K8 UI shell alone.

Chapter 5

Step-by-step: run from zero

5.1 Prerequisites

Docker, Git, optional `gh` for private repo auth. Install CLIs:

```
cd /path/to/cxr-ops-lab
./scripts/00-install-tools.sh
export PATH="$PWD/bin:$PATH"
```

5.2 Option A — Full end-to-end (recommended)

```
./scripts/15-e2e-deploy.sh
# Terraform apply -> 12-k8-ensure -> 13-argo-install -> port-forwards
    -> 14-stack-test
```

5.3 Option B — Manual layers

```
# SW.5 / cluster
cd terraform && terraform init && terraform apply && cd ..
# SW.1 + load
./scripts/02-build-and-load.sh
# SW.4
./scripts/05-helm-install.sh
# SW.8
export CXR_ARGO_REPO_URL=https://github.com/UdonsiKalu/cxr-ops-lab.git
./scripts/13-argo-install.sh
# Access
kubectl port-forward -n cxr-ui svc/cxr-ui 8081:3000 --address=127.0.0.1
kubectl port-forward -n argocd svc/argocd-server 8083:443 --address
    =127.0.0.1
```

5.4 Argo CD login

```
kubectl -n argocd get secret argocd-initial-admin-secret \
  -o jsonpath='{.data.password}' | base64 -d; echo
# User: admin  |  URL: https://localhost:8083
```

5.5 Verify

```
./scripts/14-stack-test.sh
kubectl get application cxr-ui -n argocd
helm list -n cxr-ui
curl -s -o /dev/null -w '%{http_code}\n' http://127.0.0.1:8081/
```

Chapter 6

Helm and GitOps workflow

6.1 Helm role

Helm is the **package manager**: `helm/cxr-ui/values.yaml` sets image tag, replicas, env; templates render Deployment/Service/Namespace.

6.2 Day-to-day change loop

1. Edit `helm/cxr-ui/values.yaml` (or templates).
2. `git commit && git push` to `cxr-ops-lab`.
3. Argo Application `cxr-ui` syncs (auto-sync enabled).
4. Confirm in Argo UI: **Synced / Healthy**.
5. Open `http://localhost:8081` (if port-forward running).

6.3 What is not "set and forget"

- kind cluster may be stopped to save CPU.
- Image `cxr-ui:local` must be rebuilt and `kind load` after app changes.
- Port-forwards are not in Kubernetes — restart or use `cxr-k8-forward.service`.

Chapter 7

Script reference

Script	Purpose
`00-install-tools.sh`	Install kind, kubectl, helm, terraform, argocd into `bin/`.
`01-kind-cluster.sh`	Create `cxr-lab` if missing.
`02-build-and-load.sh`	Build `cxr-ui:local` from rehearsal UI; load into kind.
`03-k8-up.sh`	One-shot: cluster + build/load + Helm.
`05-helm-install.sh`	`helm upgrade -install cxr-ui`.
`12-k8-ensure.sh`	Idempotent: start kind, load image, Helm upgrade.
`13-argo-install.sh`	Install Argo CD; repo secret; Application from GitHub.
`14-stack-test.sh`	Integration test (terraform, helm, argo, HTTP probes).
`15-e2e-deploy.sh`	Orchestrates full stack.

Chapter 8

Adapting to another deployment

8.1 Checklist

1. Fork or clone `cxr-ops-lab`; set `CXR_UI_SRC` to your app path.
2. Replace `kind` with cloud cluster OR keep kind for dev.
3. Use a **container registry** (GHCR/ECR): build, push, set `values.yaml` `image.repository` and `tag`.
4. Update Argo `CXR_ARGO_REPO_URL` and private repo credentials.
5. Replace port-forward with **Ingress** or cloud LoadBalancer.
6. Wire SQL/Qdrant via in-cluster services, ExternalName, or env (see M4.8 diagram).
7. Optional: add GitHub Actions job for `helm upgrade` (push-based CD) instead of or alongside Argo.

8.2 Syllabus mapping

Syllabus	Artifact
SW.1	`Dockerfile`, `cxr-ui:local`
SW.2	Compose (parallel, :3000)
SW.3	`k8s/`, kind, :8081
SW.4	`helm/cxr-ui/`
SW.5	`terraform/`
SW.8	Argo CD + `k8s/argocd/`

Chapter 9

Troubleshooting

- **:8081 refused** — start kind node; run `12-k8-ensure.sh`; start port-forward.
- **Argo OutOfSync / private repo** — ensure `repo-cxr-ops-lab` secret + `gh auth token`.
- **Helm adopt error** — `05-helm-install.sh` adopts namespace from raw kubectl apply.
- **Analyze fails on :8081** — expected; use :8251 or :3000 Compose for full analyze.
- **ImagePullBackOff** — kind needs `kind load docker-image cxr-ui:local`.

Chapter 10

Evidence and further reading

- evidence/SW3-k8-evidence.md, SW4-helm-verify-2026-05-28.md, SW8-e2e-stack-2026-05-28.md
- docs/K8-DEPLOY.md, docs/ADR-SW5-terraform-kind.md
- GitHub: https://github.com/UdonsiKalu/cxr-ops-lab